Metrics (how good model is?)

Not used for training ,only for evaluation

1.Accuracy metrics

Accuracy → % of correct predictions
Binary Accuracy → For yes/no classification
Categorical Accuracy → For multi-class (one-hot)
Sparse Categorical Accuracy → For integer labels
Top-1 Accuracy → Exact prediction match
Top-5 Accuracy → Correct class in top 5
Balanced Accuracy → Handles class imbalance

2. Confusion Matrix-Based Metrics

True Positives (TP) → Correct positive predictions

True Negatives (TN) → Correct negative predictions

False Positives (FP) → Wrong positive predictions

False Negatives (FN) → Missed positives

3. Probabilistic / Loss-Based Metrics

Binary Crossentropy (Log Loss)

Categorical Crossentropy

Sparse Categorical Crossentropy

Kullback-Leibler Divergence (KL Divergence)

Poisson Loss

Hinge Loss

Squared Hinge Loss

Categorical Hinge Loss

Focal Loss (handles imbalance)

Log-Cosh Loss

4. Regression Metrics

Mean Squared Error (MSE) → Average of squared differences between predicted and actual values (penalizes large errors more).

Root Mean Squared Error (RMSE) → Square root of MSE, giving error in the same units as the target.

Mean Absolute Error (MAE) → Average of absolute differences between predictions and actual values (easy to interpret).
Mean Absolute Percentage Error (MAPE) → Average percentage error between predicted and actual values.
Mean Squared Logarithmic Error (MSLE) → Measures error on a log scale, useful when dealing with large value ranges.
R^2 Score (Coefficient of Determination) → Indicates how well the model explains the variance in the data (1 = perfect fit).
Cosine Similarity → Measures similarity between predicted and actual vectors based on direction, not magnitude.

5.Classification Curve Metrics

ROC Curve
AUC (Area Under Curve)
Precision-Recall Curve (PR Curve)

6. Precision / Recall Family

Precision → Correctness of positive predictions
Recall (Sensitivity) → Ability to find positives
Specificity → Ability to detect negatives
F1 Score → Balance of precision & recall
Fβ Score → Weighted F1 (focus on recall/precision)
Precision@K → Precision at top K results
Recall@K → Recall at top K
Sensitivity at Specificity
Specificity at Sensitivity

7. Segmentation Metrics

IoU (Intersection over Union)
Mean IoU
Dice Coefficient
Binary IoU
One-Hot IoU
Hausdorff Distance

8.